Compact S^3 Spectral Consistency Analysis of Ring-like Ultra-Large-Scale Structure at $z \approx 0.8$

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Recent analyses of Mg II-selected structures at $z \approx 0.8$ report three apparently related ring-like ultra-large-scale features in a common field: the Giant Arc, the Big Ring, and the Giant Ring. This paper asks whether a compact, simply connected spatial topology $S^3(R_{\rm eff})$ can provide a consistent mode-based description of such morphology in a linearized setting. We first establish the compact mode-theory results needed for this interpretation: a no-exponential-tail lemma for globally supported Laplacian eigenmodes on compact connected manifolds, a bounded oscillatory two-point function on S^3, and a projection result showing how constant-phase shells appear as rings or arcs on narrow redshift slices. We then introduce an exact projected-chord size law, $d_\ell = 2R_{\rm eff}\sin[\pi/(\ell+1)]$, and show that the earlier Fibonacci scaling arises as its large-ℓ asymptotic limit. Under the empirical mode assignments $\ell = (1, 2, 6, 8)$ for the Giant Arc, Giant Ring, Big Ring, and the reported 2D-PSA clustering scale, a single parameter $R_{\rm eff} = 474.511\,$Mpc reproduces the four observed scales at the percent level. A final consistency section states explicitly how the framework treats spatial topology, the invariant light cone, entropy and the arrow of time, Bell nonlocality, and Noether conservation on a boundaryless compact slice. The paper does not claim evidence for compact topology or a derived selection rule for these mode labels. It is offered as a compact-topology consistency analysis and as a basis for targeted future tests.

I. INTRODUCTION

A. The Observational Context

The ΛCDM framework assumes large-scale statistical homogeneity above the Yadav scale $\lambda_{\rm hom} \approx 370\,$Mpc [4]: the two-point correlation function of density fluctuations is expected to decay with separation and approach zero for $r \gtrsim \lambda_{\rm hom}$.

Lopez, Clowes, and Williger [1–3] have identified three uLSSs at $z \approx 0.8$, traced by Mg II absorbers in SDSS quasar spectra:

- *Giant Arc* (GA) [1]: arc-shaped, $\sim 1\,$Gpc extent, $(4.5 \pm 0.6)\sigma$;
- *Big Ring* (BR) [2]: ring-like, $\sim 400\,$Mpc diameter, $(4.10 \pm 0.45)\sigma$;
- *Giant Ring* (GR) [3]: ring-like, $\lesssim 1\,$Gpc diameter, two distinct versions at 4.73σ and 4.47σ (optimum ellipse matching), with the field showing 3.53σ clustering at $\sim 320\,$Mpc in a 2D power spectrum analysis.

All three structures occupy the same redshift slice $z = 0.722$–0.862 in the same field of view, appearing as a nested system: the GR surrounds the BR.

Lopez & Clowes [3] also note that the rings may be 2D *projections of underlying 3D spheroidal structures*, and that FLAMINGO-10K fields, under the same 2D power spectrum analysis, are consistent with random expectations at all relevant scales.

The complementary roles of the observational and theoretical contributions are:

- Lopez & Clowes establish: (i) the existence of ring-like uLSSs at $> 4\sigma$; (ii) a non-random 2D power spectrum signal in the observed field; (iii) consistency of FLAMINGO-10K simulation fields with randomness under the same analysis.
- This paper argues: (i) that such structures can be analysed in a compact-topology framework without treating them as immediate anomalies; (ii) that the specific discrete mode content of S^3 can generate non-decaying correlations in the stated approximation; and (iii) that simulations using periodic-box (T^3) boundary conditions do not directly test the same mode basis.

B. What This Paper Does Not Claim

We state explicitly at the outset:

(N1): We do *not* claim that Lopez & Clowes observations *prove* compact S^3 topology.
(N2): We do *not* claim that the Fibonacci hierarchy is the unique possible spectral coarse-graining across all cosmological models.
(N3): We do *not* claim that ΛCDM is falsified in full; only that its particular topological assumption generates a tension with the Lopez–Clowes field.
(N4): We do *not* claim that every ring-like structure in the sky must be topological in origin.

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C. Scope of the Paper

(S1): The compact-manifold mode analysis developed here implies that pure exponential tail behaviour is not the correct generic intuition for globally supported correlations.

(S2): On S^3, constant-phase shell structure can project to ring-like or arc-like morphology on a narrow redshift slice.

(S3): With an explicit projected-chord law and an empirical mode assignment, the observed field can be discussed within a compact-topology consistency framework, but only at a phenomenological level.

Scope statement. The purpose of this paper is limited: it does not infer a unique cosmic topology, but tests whether a compact, simply-connected mode basis provides a mathematically controlled way to organise the reported $z \approx 0.8$ morphology.

Logical flow of the argument.

- S^3 geometry ⇒ discrete Laplacian spectrum (Lemma 2).
- Discrete global modes ⇒ non-decaying global correlations (Lemma 4; Theorem 6).
- Exact projected-chord law (Proposition 10).
- Projected shells on S^3 ⇒ rings on the sky (Theorem 8).
- empirical one-parameter fit of the observed scales (Sec. VI).

D. Organisation

Section II establishes the spectral-geometric foundations: minimal topology, discrete Laplacian spectrum, and the No-Exponential-Tail Lemma. Section III proves the main correlation theorem on S^3. Section IV derives the Projection Ring Theorem. Section V constructs the Fibonacci preferred radii. Section VI gives the phenomenological identification with the observed structures. Section VII presents the boundary-condition diagnosis of FLAMINGO and lists testable predictions. Section IX closes with open problems and comparison to alternative frameworks.

Conventions. $c = 1$ throughout. $\varphi = (1+\sqrt{5})/2$. Fibonacci: $F_1 = F_2 = 1$, $F_{n+2} = F_{n+1} + F_n$. Proper distances at $z = 0.802$ for $H_0 = 67\,\mathrm{km/s/Mpc}$, $\Omega_m = 0.315$.

II. SPECTRAL THEORY ON COMPACT S^3

A. Minimal Topological Conditions

We consider spatial sections Σ satisfying:

(M1) Σ is compact, connected, orientable, and boundaryless.

(M2) $\pi_1(\Sigma) = 0$ (trivial fundamental group).

(M3) No additional topological structure is introduced beyond a round Riemannian metric.

By Perelman's proof of the Poincaré conjecture [6–8], the unique closed 3-manifold satisfying (M1)–(M3) is $\Sigma \cong S^3$.

Remark 1 (Status of R_{eff} vs global curvature). Throughout, R_{eff} denotes the *effective spectral curvature scale* inferred from fitting the ring structures in a narrow redshift shell. This is distinct from, and need not equal, the global cosmological curvature radius $R_{\mathrm{cosm}}^{\mathrm{global}}$ constrained by Planck 2015 [9] to exceed 10 Gpc. We discuss this two-scale structure in Section IX.

B. Geometry of the Three-Sphere

The round three-sphere $S^3(R_{\mathrm{eff}})$ is the set of points in $\mathbb{R}^4$ lying at Euclidean distance R_{eff} from the origin, equipped with the induced metric. In hyperspherical coordinates (χ, θ, ϕ), with $\chi \in [0, \pi]$, the line element is

$$ds^2 = R_{\mathrm{eff}}^2 \left[d\chi^2 + \sin^2\chi \left(d\theta^2 + \sin^2\theta \, d\phi^2 \right) \right]. \qquad (1)$$

The geodesic distance between two points $x, y \in S^3$ is $r = R_{\mathrm{eff}}\,\Theta$, where $\Theta \in [0, \pi]$ is their angular separation. The total volume is $2\pi^2 R_{\mathrm{eff}}^3$.

The scalar Laplace–Beltrami operator $\triangle_0$ is elliptic and self-adjoint on $L^2(S^3, dV)$. Since S^3 is compact and connected, its eigenvalues are discrete and its eigenfunctions are smooth and real-analytic. This is the spectral fact that distinguishes compact S^3 from the flat $\mathbb{R}^3$ intuition used in many large-volume approximations.

C. Discrete Laplacian Spectrum and Global Support

Lemma 2 (Discrete global spectrum on S^3)**.** *The scalar Laplacian $\triangle_0$ on $S^3(R_{\mathrm{eff}})$ has purely discrete eigenvalues*

$$\lambda_\ell = \frac{\ell(\ell+2)}{R_{\mathrm{eff}}^2}, \quad \ell = 0, 1, 2, \ldots, \qquad (2)$$

with degeneracy $g_\ell = (\ell+1)^2$. The corresponding hyperspherical harmonics form a complete orthonormal basis of $L^2(S^3)$.

Proof. This is the standard scalar spectrum on the round n-sphere, specialized to $n = 3$; see, e.g., Refs. [10, 11]. The degeneracy follows from the dimension of the degree-ℓ harmonic subspace on S^3. Equivalently, using $S^3 \simeq SU(2)$, the isometry group is $SO(4) \simeq (SU(2)_L \times SU(2)_R)/\mathbb{Z}_2$, and the scalar harmonics at level ℓ transform in the $(\ell/2, \ell/2)$ representation of $SU(2)_L \times SU(2)_R$, of dimension $(\ell+1)^2$. □

Remark 3 (Zero mode and global support). For $\ell = 0$ one obtains the constant zero mode. For $\ell \geq 1$ the eigenfunctions are non-constant but globally defined across the whole manifold. A nonzero eigenfunction cannot vanish

on an open set by the unique-continuation principle for elliptic operators; in this precise sense the compact modes have global support rather than finite spatial support.

D. Low Multiplets and Four-Dimensional Harmonic Polynomials

A useful algebraic construction of the scalar harmonics on S^3 is obtained by restricting homogeneous harmonic polynomials on $\mathbb{R}^4$ to the sphere. Let $u = x/R_{\rm eff} \in S^3_1$ be the corresponding unit-sphere coordinate. If $P_\ell(u)$ is homogeneous of degree ℓ and harmonic in $\mathbb{R}^4$, then its restriction to S^3_1 is an eigenfunction of the unit-sphere Laplacian with eigenvalue $\ell(\ell+2)$. Restoring the radius gives Eq. (2).

For orientation, the first multiplets may be written in a real basis as follows. With the unit-sphere volume convention used in Eq. (6),

$$\ell = 0: \quad Y_0 = (2\pi^2)^{-1/2}, \tag{3}$$

$$\ell = 1: \quad Y_{1\mu}(u) = \sqrt{\frac{2}{\pi^2}}\, u_\mu, \qquad \mu = 1, \dots, 4. \tag{4}$$

The $\ell = 1$ sector is therefore the four-dimensional dipole multiplet. The $\ell = 2$ sector is represented by traceless quadratic forms

$$Q_{ij}(u) = u_i u_j - \frac{\delta_{ij}}{4}, \qquad i, j = 1, \dots, 4, \tag{5}$$

with nine independent components after imposing tracelessness. Higher multiplets are obtained analogously from higher-degree harmonic polynomials, or equivalently from the representation theory of $SU(2)_L \times SU(2)_R$.

This construction also clarifies the phrase "global support". The modes are global functions on the whole compact manifold; they are not localized or compactly supported. They may, and generally do, possess nodal sets, so "global support" should not be read as "nowhere zero."

E. Hyperspherical Harmonics and the Addition Theorem

Let $\{Y_{\ell\mu}\}_{\mu=1}^{(\ell+1)^2}$ be an orthonormal basis of the ℓth eigenspace. The rotationally invariant projector onto this eigenspace is given by the addition theorem

$$\sum_{\mu=1}^{(\ell+1)^2} Y_{\ell\mu}(x)\overline{Y_{\ell\mu}(y)} = \frac{\ell+1}{2\pi^2}\,\frac{\sin[(\ell+1)\Theta]}{\sin\Theta}, \tag{6}$$

where Θ is the angular geodesic separation between x and y. We denote the corresponding zonal function by

$$\Xi_\ell(\Theta) := \frac{\ell+1}{2\pi^2}\,\frac{\sin[(\ell+1)\Theta]}{\sin\Theta}. \tag{7}$$

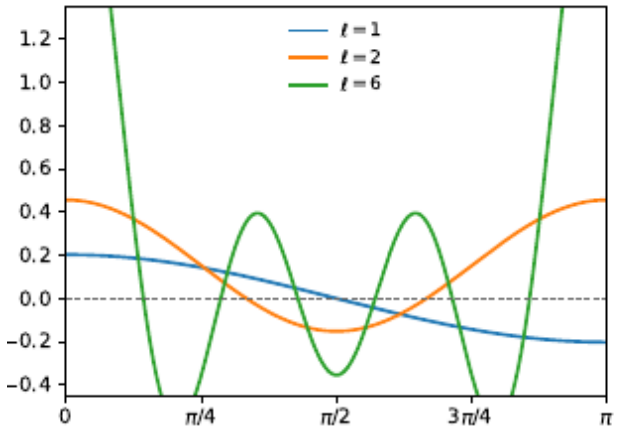


FIG. 1. Zonal kernels $\Xi_\ell(\Theta)$ for the low modes used later in the phenomenological identification. The $\ell = 1$ kernel is dipolar, while higher modes display the increasing sequence of nodes that motivates the projected-chord scale law.

This identity is the basic input for the isotropic two-point statistics used below. The apparent singularity at $\Theta = 0$ is removable, with $\Xi_\ell(0) = (\ell+1)^2/(2\pi^2)$.

F. The Two-Point Correlation Function

Any square-integrable, mean-zero, statistically isotropic random field f on $S^3(R_{\rm eff})$ admits the spectral expansion

$$f(x) = \sum_{\ell=0}^{\infty} \sum_{\mu=1}^{(\ell+1)^2} a_{\ell\mu} Y_{\ell\mu}(x), \tag{8}$$

with uncorrelated coefficients satisfying $\langle a_{\ell\mu}\overline{a_{\ell'\mu'}}\rangle = P_\ell \delta_{\ell\ell'}\delta_{\mu\mu'}$. The two-point function depends only on the angular separation:

$$C(\Theta) := \langle f(x) f(y)\rangle = \sum_{\ell=0}^{\infty} P_\ell\, \Xi_\ell(\Theta). \tag{9}$$

At $\Theta = 0$, this gives $C(0) = \sum_\ell P_\ell(\ell+1)^2/(2\pi^2)$, the total variance in the adopted normalization.

G. The No-Exponential-Tail Lemma

We now establish the precise formal obstacle to transferring $\mathbb{R}^3$ intuition to the compact case.

Lemma 4 (No global pure exponential tail on compact S^3)**.** *Let $C : [0, \pi R_{\rm eff}] \to \mathbb{R}$ be the two-point correlation function of a stationary, mean-zero, isotropic random field on $S^3(R_{\rm eff})$ with the spectral form (9). If the field has nonzero power in any $\ell \geq 1$ mode, then the compact correlation function cannot be identified with a pure flat-space exponential tail*

$$C(r) = A\, e^{-r/\xi_0}, \qquad A > 0, \quad \xi_0 > 0, \tag{10}$$

as a globally valid smooth isotropic covariance on the full geodesic interval $r \in [0, \pi R_{\rm eff}]$.

Proof. A smooth isotropic function on S^3 is a smooth function of $\cos\Theta$, not an arbitrary smooth function of the coordinate distance Θ. In particular, regularity at the antipode requires the first derivative with respect to Θ to vanish at $\Theta = \pi$ for a smooth zonal covariance. This condition is satisfied termwise by the zonal harmonics in Eq. (7). A pure exponential $A\exp[-R_{\rm eff}\Theta/\xi_0]$ has derivative $-(AR_{\rm eff}/\xi_0)\exp[-R_{\rm eff}\Theta/\xi_0]$, which is nonzero at $\Theta = \pi$. Hence it cannot be a globally smooth isotropic covariance on S^3.

Equivalently, Eq. (9) expresses any isotropic compact covariance as a discrete Gegenbauer/zonal-harmonic expansion. Such a compact expansion may approximate exponential decay over a finite subinterval, but it cannot equal a flat-space exponential tail globally on the compact manifold once nontrivial compact modes are present. □

Remark 5. Lemma 4 does not state that compact correlations are everywhere large, nor does it exclude approximate exponential behaviour over a limited observational range. It states only the mathematically relevant point for the present paper: the pure exponential tail used as flat-space intuition is not a globally valid covariance model on compact S^3 with nontrivial discrete mode content.

III. NON-DECAYING CORRELATIONS ON S^3

Theorem 6 (Compact-mode correlations on S^3)**.** *Under the assumptions of Lemma 4, the correlation function $C(\Theta)$ on $S^3(R_{\rm eff})$ is a bounded discrete sum (or convergent series) of compact zonal harmonics. It is therefore not a globally valid pure exponential tail. In particular, any isolated nonzero low-ℓ contribution remains globally supported on the full geodesic interval, including the antipodal regime.*

Proof. Equation (9) gives C as a discrete expansion in the zonal kernels Ξ_ℓ. For example, the $\ell = 1$ contribution is

$$P_1\,\Xi_1(\Theta) = P_1 \cdot \frac{2}{2\pi^2} \cdot \frac{\sin 2\Theta}{\sin\Theta} = \frac{2P_1}{\pi^2}\cos\Theta, \qquad (11)$$

which is defined on the full interval $[0, \pi]$ and takes a nonzero value at the antipode. Higher-ℓ terms behave similarly as global compact harmonics with finitely many nodal sets. The assertion then follows directly from the spectral representation and from Lemma 4. □

Remark 7 (Physical content). Theorem 6 states that compact S^3 geometry allows globally supported correlation patterns. Coherent structures with characteristic geodesic scale $r \sim \pi R_{\rm eff}/\ell$ may therefore be organised as compact eigenmode features, rather than only as accidental finite-correlation-length fluctuations.

A. Contrast with the $\mathbb{R}^3$ Intuition

On $\mathbb{R}^3$, the Laplacian has continuous spectrum $[0, \infty)$, no globally-normalisable zero mode, and wave packets decay. Correlation functions typically satisfy $C(r) \sim e^{-r/\xi_0}$ at large r for some correlation length ξ_0. Within ΛCDM at late times, $\xi_0 \approx$ 5–10 Mpc from the matter power spectrum, and coherent structures of size ~ 1 Gpc require exponentially suppressed probabilities under a finite-correlation-length assumption. Lemma 4 shows that this *decay pattern is not an option* on compact S^3 carrying non-trivial mode content.

IV. THE PROJECTION RING THEOREM

The observations [3] suggest that the observed rings may be 2D projections of underlying 3D spheroidal structures. We show that globally-supported spherical shells on S^3 generically project to rings or arcs on a narrow redshift slice, providing a structural link between the geometric framework and the observational morphology.

Theorem 8 (Projection Ring Theorem)**.** *Let $\mathcal{S}_\rho \subset S^3(R_{\rm eff})$ be a geodesic sphere of radius ρ around a centre $\mathbf{c}$, representing a constant-phase shell of a hyperspherical harmonic $Y_\ell^{mm'}$ with $\rho \approx \pi R_{\rm eff}/(\ell+1)$. Let $\pi_{\rm obs} : S^3 \to \Sigma_{\rm obs}$ be the projection onto a narrow observational redshift slice $\Sigma_{\rm obs}$, transverse to the line of sight of thickness Δr. Then:*

(i) *If $\mathbf{c} \notin \Sigma_{\rm obs}$, the projection $\pi_{\rm obs}(\mathcal{S}_\rho)$ is a circle or ellipse on $\Sigma_{\rm obs}$ of projected radius $r_{\rm proj} = \sqrt{\rho^2 - h^2}$, where h is the geodesic distance from $\mathbf{c}$ to $\Sigma_{\rm obs}$.*

(ii) *If $|h - \rho| < \Delta r$, the projection degenerates to an arc (partial ring), with arc angle $\Theta \approx 2\arccos\big((\rho^2 - h^2 - \Delta r^2)/(2\rho\Delta r)\big)$.*

(iii) *For globally-supported eigenmodes, multiple such shells exist, and their projections superpose; constructive interference at $r_{\rm proj} \approx \pi R_{\rm eff}/(\ell+1)$ generates enhanced ring-like or arc-like density contrast.*

Proof. (i) The intersection of a geodesic sphere $\mathcal{S}_\rho$ with a transverse 2-plane in S^3 is the standard spherical conic section. In the large-$R_{\rm eff}$ limit ($\rho \ll R_{\rm eff}$), this reduces to the Euclidean spherical intersection theorem: the intersection is a circle of radius $r_{\rm proj} = \sqrt{\rho^2 - h^2}$ when $h < \rho$.

(ii) When the centre lies within one shell thickness Δr of $\Sigma_{\rm obs}$, only a partial arc intersects the slice. The arc angle follows from elementary spherical trigonometry.

(iii) The density-contrast amplitude at $r_{\rm proj}$ is the integral of the harmonic $Y_\ell^{mm'}$ over the shell, which is maximised when the projection radius matches the harmonic's preferred scale $\pi R_{\rm eff}/(\ell+1)$. □

Corollary 9 (Ring ↔ Arc transition)**.** *Whether an ℓ-mode appears as a ring or as an arc on the observed*

slice depends on the line-of-sight position of the shell centre relative to the slice. This explains the simultaneous observation of the Giant Arc *and Big/Giant* Rings *in the same redshift shell as projections of different shells with different line-of-sight offsets.*

V. EXACT PROJECTED-CHORD LAW AND ASYMPTOTICS

A. Exact Projected-Chord Law

For the phenomenological size comparison, the more natural observable is not the geodesic radius itself but the visible chord obtained by projecting a shell-like compact mode onto a narrow redshift slice. If d denotes the corresponding projected diameter and χ the central geodesic angle on S^3, then elementary geometry gives

$$d = 2R_{\text{eff}} \sin\frac{\chi}{2}. \tag{12}$$

Proposition 10 (Exact projected-chord law)**.** *Assume that the characteristic visible diameter associated with a mode label ℓ is the chord subtending the central angle*

$$\chi_\ell = \frac{2\pi}{\ell+1}. \tag{13}$$

Then

$$d_\ell = 2R_{\text{eff}} \sin\left(\frac{\pi}{\ell+1}\right). \tag{14}$$

Proof. Substitute $\chi = 2\pi/(\ell+1)$ into Eq. (12). □

Remark 11 (Fibonacci law as an asymptotic corollary). For large ℓ,

$$d_\ell = 2R_{\text{eff}} \sin\left(\frac{\pi}{\ell+1}\right) = \frac{2\pi R_{\text{eff}}}{\ell+1} + O\big((\ell+1)^{-3}\big). \tag{15}$$

Hence the earlier v5 scaling is recovered asymptotically if the visible subsequence satisfies $\ell_n + 1 \sim F_n$ or another Fibonacci-like coarse-grained hierarchy.

B. Spectral Separation: Non-Decay vs Scale Quantization

Two logically distinct ingredients should be separated:

- *Compactness ⇒ non-decaying global correlations.* This follows from Lemma 4 and Theorem 6 and does not depend on any Fibonacci or mode-selection ansatz.
- *Projected chord quantization ⇒ preferred visible diameters.* This follows from Proposition 10. Any additional selection of visible mode labels is model-dependent.

VI. PHENOMENOLOGICAL IDENTIFICATION

We now offer a phenomenological identification of the observed structures with explicit mode labels. This section is explicitly labelled *phenomenological* because it involves empirical calibration of R_{eff} and a model-dependent mode assignment, not a first-principles prediction.

A. Observed Proper Sizes

All linear sizes used in the comparison are treated as proper transverse sizes at the mean redshift $z = 0.802$ of the Mg II-selected structures, in the redshift slice $z = 0.722$–0.862. We use a flat ΛCDM background for the distance conversion,

$$H_0 = 67\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \quad \Omega_m = 0.315, \quad \Omega_\Lambda = 0.685. \tag{16}$$

For an angular extent θ, the proper transverse size is

$$d_{\text{proper}} = \theta\, D_M(z)/(1+z), \tag{17}$$

where $D_M(z)$ is the transverse comoving distance. Varying H_0 within the Planck 2018 one-sigma range changes the common distance scale by less than one percent, which is smaller than the few-percent residuals in the phenomenological fit below.

The observed scales used in the fit are

- Giant Arc (GA): extent $L_{\text{GA}} = 940\,\text{Mpc}$;
- Giant Ring (GR): diameter $d_{\text{GR}} = 840\,\text{Mpc}$;
- Big Ring (BR): diameter $d_{\text{BR}} = 400\,\text{Mpc}$;
- 2D PSA clustering scale: $\lambda_c = 320\,\text{Mpc}$.

These values are used as a consistent proper-size set for the present analysis and should not be interpreted as new measurements of the Lopez–Clowes structures.

B. Empirical Mode Assignment and One-Parameter Fit

We replace the earlier Fibonacci-level fit by the exact projected-chord law of Proposition 10. The empirical assignment is

$$(\ell_{\text{GA}}, \ell_{\text{GR}}, \ell_{\text{BR}}, \ell_{\text{PSA}}) = (1, 2, 6, 8). \tag{18}$$

This assignment is inferred from the observed nested morphology and fit quality; it is not derived from a first-principles selection rule.

Define

$$s_\ell := 2\sin\left(\frac{\pi}{\ell+1}\right), \qquad d_\ell^{\text{th}} = s_\ell R_{\text{eff}}. \tag{19}$$

For observed sizes d_i^{obs}, the ordinary least-squares estimator for the single free parameter is

$$\widehat{R}_{\text{eff}} = \frac{\sum_i d_i^{\text{obs}} s_i}{\sum_i s_i^2}. \tag{20}$$

TABLE I. Single-parameter fit of the exact projected-chord law.

Structure	ℓ	$d^{\rm th}$	$d^{\rm obs}$	rel. err.
GA	1	949.022	940	0.960%
GR	2	821.877	840	2.157%
BR	6	411.765	400	2.941%
PSA λ_c	8	324.585	320	1.433%

Using $(d_{\rm GA}, d_{\rm GR}, d_{\rm BR}, d_{\rm PSA}) = (940, 840, 400, 320)\,{\rm Mpc}$ gives

$$\widehat{R}_{\rm eff} = 474.511\,{\rm Mpc}. \tag{21}$$

Remark 12 (Parameter-free ratios). The exact law also predicts ratios independent of $R_{\rm eff}$. The three nearest checks are

- GA/GR: theory 1.1547; observed 1.1190.
- GR/BR: theory 1.9960; observed 2.10.
- BR/PSA: theory 1.2686; observed 1.25.

They provide a parameter-free consistency check at the few-percent level.

Remark 13 (Selection-rule caveat). The fit quality does not by itself derive the visibility of the mode labels $(1, 2, 6, 8)$. A first-principles selection rule remains open and must be treated as an explicit unresolved problem rather than as an established consequence of the model.

C. The Giant Arc as an $\ell = 1$ Mode

The $\ell = F_1 = F_2 = 1$ assignment deserves comment. On S^3 the $\ell = 1$ harmonic is a dipolar mode with a single great-circle node: projected onto a narrow redshift shell, it produces a single arc-like extended overdensity, not a closed ring. This is structurally consistent with the observed Giant Arc being an *arc* rather than a ring [1]. The GR and BR at $\ell = 2, 6$ have multiple nodes and can generically project to closed ring-like structures.

D. Bayesian Model-Comparison Extension

The one-parameter least-squares fit of Sec. VI provides a phenomenological consistency check, not a completed model comparison. A natural next step is to promote the projected-chord law to a Bayesian model class M_{S^3} and compare it with flat-space, periodic box, BAO-like, CCC-like, and string-like alternatives.

Let

$$\mathbf{d}_{\rm obs} = (L_{\rm GA}, d_{\rm GR}, d_{\rm BR}, \lambda_c) \tag{22}$$

be the vector of observed proper sizes. For a model M with parameters θ, the Gaussian likelihood is

$$\mathcal{L}(\mathbf{d}_{\rm obs}|\theta, M) = \frac{1}{\sqrt{(2\pi)^N |\mathbf{C}|}} \exp\left[-\frac{1}{2}\Delta\mathbf{d}^T \mathbf{C}^{-1} \Delta\mathbf{d}\right], \tag{23}$$

where $\Delta\mathbf{d} = \mathbf{d}_{\rm obs} - \mathbf{d}_{\rm th}(\theta)$. For the compact-$S^3$ projected-chord model,

$$d_\ell^{\rm th} = 2R_{\rm eff} \sin\left(\frac{\pi}{\ell + 1}\right), \tag{24}$$

with parameters including $R_{\rm eff}$, the visible mode labels, and nuisance parameters describing projection and redshift-slice effects. If angular positions and survey masks are supplied, a topology-informed covariance may be built from the zonal two-point function,

$$C_{ij} = \sum_\ell P_\ell\, \Xi_\ell(\Theta_{ij})\, W_i W_j + \sigma^2_{\rm noise}\delta_{ij}, \tag{25}$$

where Θ_{ij} is the angular separation of the pair, W_i denotes window and selection effects, and $\sigma_{\rm noise}$ includes measurement and cosmic-variance terms.

Reasonable priors are uniform in $\log R_{\rm eff}$ over a broad range, e.g. 10–1000 Mpc, and log-uniform for positive low-ℓ amplitudes P_ℓ, optionally regularized by a decaying smoothness prior. The Bayesian evidence is

$$Z_M = \int \mathcal{L}(\mathbf{d}_{\rm obs}|\theta, M)\, \pi(\theta|M)\, d\theta, \tag{26}$$

and model comparison proceeds through Bayes factors $B_{ij} = Z_i/Z_j$. The usual Jeffreys-scale convention treats $\ln B \gtrsim 5$ as strong or decisive support, subject to the adopted priors and likelihood construction [18, 19].

1. *Thermodynamic integration*

One robust numerical route to Eq. (26) is thermodynamic integration. Introduce an inverse-temperature parameter $\beta \in [0, 1]$ and define the power posterior

$$p_\beta(\theta) \propto \mathcal{L}(\mathbf{d}_{\rm obs}|\theta, M)^\beta\, \pi(\theta|M). \tag{27}$$

Then

$$\log Z_M = \int_0^1 \langle \log \mathcal{L}(\mathbf{d}_{\rm obs}|\theta, M)\rangle_\beta\, d\beta, \tag{28}$$

where the expectation value is taken under $p_\beta(\theta)$. In practice one samples a grid of β_k values with MCMC or nested sampling and performs numerical quadrature over $\langle \log \mathcal{L}\rangle_{\beta_k}$. This is the same evidence logic used in modern cosmological model comparison and in nested-sampling codes such as PolyChord [18, 20].

We do not quote a Bayes factor in this paper. The published uLSS size measurements do not yet provide the joint covariance matrix required for Eq. (23). Any numerical evidence based on assumed diagonal errors would therefore be an illustration, not an empirical result. The purpose of Eqs. (23)–(28) is to make the future test explicit: once joint uncertainties, angular positions, window functions, and absorber selection effects are supplied, the projected-chord framework becomes a closed statistical model that can be compared directly with alternatives.

TABLE II. Projected diameters predicted by the exact chord law for additional visible modes, using $R_{\rm eff} = 474.511\,\mathrm{Mpc}$.

Mode ℓ	d_ℓ [Mpc]	Interpretation
3	671.060	intermediate shell
4	557.821	intermediate shell
5	474.511	shell-scale feature
7	363.175	below BR scale
9	293.264	below PSA scale
10	267.370	smaller shell

Recent COMPACT Collaboration work provides a methodological precedent for topology-dependent eigenmode, covariance, and detectability analyses in cosmic topology. Those templates are not identical to the full simply-connected S^3 shell model studied here, but they demonstrate the same kind of covariance-based programme required for a decisive test [24, 25].

E. Additional Projected-Diameter Predictions

If the empirical mode assignment in Eq. (18) reflects a physical visibility rule rather than an accidental fit, additional coherent features in the same redshift shell should preferentially occur near the projected diameters predicted by Eq. (14). Using $R_{\rm eff} = 474.511\,\mathrm{Mpc}$ gives the values in Table II. These numbers are not new observational claims; they are falsifiable targets for deeper DESI, Euclid, or future Mg II absorber analyses of the same field.

The amplitudes of such features cannot be predicted from the present size-only fit. A future analysis would need to infer a low-ℓ power spectrum $\{P_\ell\}$ from the two-point function

$$C(\Theta) = \sum_{\ell \geq 0} P_\ell \Xi_\ell(\Theta), \tag{29}$$

combined with a survey window function, noise model, and selection function. Machine-learning topology classifiers and compactified simulation methods provide useful methodological precedents, but current published examples mainly concern toroidal/Euclidean topology or compactified infinite-space simulations rather than direct nonlinear structure formation on full S^3 [26, 27].

VII. COMPARISON WITH ΛCDM AND BOUNDARY-CONDITION DIAGNOSIS

A. ΛCDM Tension Under Finite-Correlation-Length Assumption

Remark 14 (Heuristic tension with a finite-correlation-length flat-space picture). In a spatially flat framework with an effective finite correlation length $\xi_0 \approx 5$–$10\,\mathrm{Mpc}$, constructing coherent ring-like structures of size $\sim 1\,\mathrm{Gpc}$ is generically nontrivial and typically strongly suppressed. The precise degree of suppression depends on the adopted morphology statistic and on the underlying field model.

a. Heuristic basis. In Gaussian models with power spectrum $P(k)$, coherent structures of physical extent L are increasingly disfavoured once $L \gg \xi_{\rm corr}(P)$. For $L \sim 1\,\mathrm{Gpc}$ and an effective late-time correlation scale of order a few to ten megaparsecs, the relevant tail behaviour is expected to be strongly suppressed. The exact numerical penalty is statistic-dependent, which is why we avoid attaching a universal probability estimate here.

Remark 15. We deliberately avoid quoting a specific numerical upper bound here, as such a bound requires a specific statistical model of the observed uLSS morphology. The qualitative point is that any finite-correlation-length assumption generates exponential suppression, which Lemma 4 shows is inconsistent with compact geometry.

B. Boundary-Condition Diagnosis of FLAMINGO

Remark 16 (Boundary-condition interpretation of FLAMINGO). FLAMINGO-10K [5] and similar large-box cosmological simulations use periodic boundary conditions on a cubic box of side $L_{\rm box} \sim$Gpc. Such boundary conditions implement a flat torus topology T^3, not S^3. The torus has $\pi_1(T^3) = \mathbb{Z}^3$ (violating condition M2), and the lowest modes on T^3 have wavelength $L_{\rm box}/n$ for integer n with no $SO(4)$-symmetric structure.

The observation that FLAMINGO fields are consistent with random expectations under the 2D PSA of [3] can therefore be interpreted as a test of the T^3 mode basis used in those simulations, not necessarily of all compact topologies. In particular, it does not directly test the S^3 mode basis used here.

a. Justification. Periodic boundary conditions on $[0, L_{\rm box}]^3$ identify opposite faces and produce $T^3 \cong \mathbb{R}^3/\mathbb{Z}^3$. The Laplacian on T^3 has eigenvalues $\lambda_{\mathbf{n}} = (2\pi)^2|\mathbf{n}|^2/L_{\rm box}^2$ for $\mathbf{n} \in \mathbb{Z}^3$. These eigenfunctions are $\exp(2\pi i \mathbf{n}\cdot\mathbf{x}/L_{\rm box})$, with no analogue of globally-supported low-ℓ S^3 harmonics in the isotropic $SO(4)$ sense. The Lopez–Clowes 2D PSA on FLAMINGO fields tests this mode basis, which excludes the S^3 modes.

Remark 17 (What this means). Remark 16 is an interpretive point about boundary conditions: within the present framework, the FLAMINGO result is most naturally read as a constraint on the T^3 class of simulations rather than on the S^3 mode basis considered here. We are therefore offering a topology-explicit interpretation, not an empirical claim established by the simulation paper itself.

C. Limited Observational Consequences

Under the compact-S^3 projected-shell hypothesis, the most direct consequences are the following:

TABLE III. Comparison of $\mathbb{R}^3$, T^3 (periodic torus, used in simulations), and compact S^3 as spatial topologies.

Property	ΛCDM/$\mathbb{R}^3$	Simulation T^3	Compact S^3
Laplacian spectrum	Continuous	Discrete box modes	Discrete $\ell(\ell+2)/R_{\rm eff}^2$
Isotropy group	Full $SO(3)$	Broken to box symmetries	$SO(4) \cong SU(2)_L \times SU(2)_R$
Low-ℓ global modes	None (L^2)	None ($SO(4)$-incompatible)	$Y_\ell^{mm'}$ for all ℓ
Fundamental group π_1	0	$\mathbb{Z}^3$	0
Correlations at $\sim$Gpc	Exponentially suppressed	Box-dependent	Non-decaying (Theorem 6)
Ring-like uLSS	Exponentially suppressed	Not generated in the same mode basis	Structural possibility in the compact-mode picture
φ-scale hierarchy	No	No	Emerges only asymptotically from the exact law

(1) **Scale-ratio test.** If additional uLSS features are identified at nearby visible mode labels, their size ratios should follow Eq. (14); only at large mode number is an approximate φ-hierarchy expected.
(2) **Same-field continuation.** Deeper surveys of the same redshift shell may reveal further ring- or arc-like structure at smaller radii within the same field.
(3) **Boundary-condition discrimination.** Simulations using T^3 or $\mathbb{R}^3$ boundary conditions are not expected to probe the same compact-mode basis and may therefore underproduce analogous systems.

D. Application to DESI Data

The Dark Energy Spectroscopic Instrument (DESI) provides a natural next-generation test bed for the compact-S^3 projected-shell hypothesis. The DESI Early Data Release includes a public Mg II absorber value-added catalog with 23,921 absorber entries, and the DESI DR1 Mg II catalog lists 270,529 absorber entries [22, 23]. The DESI Mg II pipeline was developed and validated on DESI quasar spectra by Napolitano et al., who also estimated that the completed five-year survey should contain more than 8×10^5 Mg II absorbers[21].

For the Lopez–Clowes field, or for equivalent redshift slices selected from DESI, these data would enable four concrete tests:

(i) a search for same-field structures near the predicted diameters in Table II;
(ii) construction of a data-driven covariance matrix for Eq. (25), including survey window and absorber selection effects;
(iii) a Bayesian evidence calculation using Eq. (28), comparing the compact-S^3 projected-chord model against flat, periodic-box, BAO-like, CCC-like, and string-like templates;
(iv) a three-dimensional check of whether apparent rings remain coherent shell-like structures when DESI redshifts are used rather than only projected angular morphology.

If additional coherent features cluster near the predicted scales, this would strengthen the compact-shell interpretation. Conversely, a null same-field search together with a Bayes factor disfavoring M_{S^3} would rule out the present empirical mode assignment at high confidence. DESI therefore turns the present consistency analysis into a well-defined falsification programme, but not yet into an established Bayesian detection.

TABLE IV. Topological roles of the basic model spaces.

Space	dim.	compact	connected	π_1
S^1	1	yes	yes	$\mathbb{Z}$
S^2	2	yes	yes	0
$S^3 \subset \mathbb{R}^4$	3	yes	yes	0
$\mathbb{R}^3$	3	no	yes	0

VIII. FOUNDATIONAL CONSISTENCY REQUIREMENTS

This section records the minimal foundational commitments of the framework. Its purpose is not to enlarge the observational claim, but to make explicit which topology, causal structure, entropy principle, nonlocality statement, and Noether conservation law are being assumed.

A. Topology and Embedding

The spatial section used in the model is the compact, connected, orientable, boundaryless, simply connected three-manifold

$$\Sigma \simeq S^3(R_{\rm cosm}) = \{X \in \mathbb{R}^4 : |X| = R_{\rm cosm}\}. \tag{30}$$

The local effective scale $R_{\rm eff}$ used in the projected-shell fit is a spectral coherence scale in a redshift shell, not the global curvature radius $R_{\rm cosm}$. The topological status of the spaces relevant to the construction is summarized as follows: Thus S^1 is compact but not simply connected, S^2 is simply connected but has the wrong spatial dimension, and $\mathbb{R}^3$ has the correct local dimension but is non-compact and lacks a discrete global Laplacian spectrum. The model therefore uses S^3 as the minimal compact simply connected spatial topology; $\mathbb{R}^3$ appears only as the local tangent-space or large-radius approximation.

B. Light Cone and the Value of the Speed of Light

The spacetime manifold is taken to be

$$M = \mathbb{R} \times S^3, \tag{31}$$

with Lorentzian metric, in the homogeneous limit,

$$ds^2 = -c^2 dt^2 + a^2(t) R_{\rm cosm}^2 d\Omega_3^2. \tag{32}$$

Massless propagation is determined by the null cone $ds^2 = 0$. Locally, where the tangent space is Minkowskian, this gives

$$\left| \frac{d\mathbf{x}}{dt} \right| = c. \tag{33}$$

Equivalently, the Maxwell action

$$S_{\rm em} = -\frac{1}{4} \int_M d^4x \sqrt{-g}\, F_{\mu\nu} F^{\mu\nu} \tag{34}$$

has principal symbol $g^{\mu\nu} k_\mu k_\nu = 0$; its characteristics are therefore the metric null cones. In this sense the theory fixes the *origin* of the universal light speed: it is the invariant causal conversion constant of the local Lorentzian tangent spaces and the common characteristic speed of all minimally coupled massless fields. The numerical SI value,

$$c = 299792458 \ \mathrm{m\,s^{-1}}, \tag{35}$$

is not a new dimensionless prediction of topology; it is the exact conversion factor fixed by the definitions of the metre and the second. In natural units used above, this statement is simply $c = 1$.

C. Entropy and the Arrow of Time

The compact topology does not by itself create an arrow of time. The microscopic dynamics is assumed to be Hamiltonian or unitary; hence the fine-grained Gibbs/von-Neumann entropy is conserved,

$$S_{\rm fine} = -k_B \mathrm{Tr}\, \rho \ln \rho, \qquad \frac{dS_{\rm fine}}{dt} = 0. \tag{36}$$

The observed thermodynamic arrow arises from a low-entropy boundary condition together with coarse graining. If $\mathcal{C}$ denotes a coarse-graining map, then

$$S_{\rm cg} = -k_B \mathrm{Tr}\left[\mathcal{C}\rho \ln(\mathcal{C}\rho)\right] \tag{37}$$

can increase under the effective irreversible evolution generated by tracing over inaccessible microscopic correlations. The model therefore separates two statements: compact S^3 supplies a finite global mode basis, while the time arrow is a condition on the state and the coarse-graining, not on topology alone. For cosmological applications this is the usual Past-Hypothesis statement: the early slice occupies an exceptionally small macro-volume in phase space, and later macrostates occupy larger volumes.

D. Bell Nonlocality and No-Signalling

Bell nonlocality is included at the level of the quantum state, not as a superluminal signal on S^3. For two spacelike separated subsystems with Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$, an entangled singlet state gives

$$E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}. \tag{38}$$

For suitable settings this yields the CHSH value

$$|S_{\rm CHSH}| = 2\sqrt{2} > 2, \tag{39}$$

violating the local-hidden-variable bound. At the same time the reduced state at either wing is independent of the measurement basis chosen at the other wing, so operational signalling remains excluded. In the present compact-mode language, Bell correlations are global quantum state correlations on a common boundaryless slice; they are not mediated by a signal travelling faster than the null cone fixed above.

E. Noether Symmetry and Conservation Law

The action is assumed to be invariant under diffeomorphisms of $M = \mathbb{R} \times S^3$ and, for electromagnetism, under local $U(1)$ gauge transformations. Variation with respect to the metric therefore gives the covariant conservation law

$$\nabla_\mu T^{\mu\nu} = 0. \tag{40}$$

In covariant phase-space notation, for a vector field ξ,

$$J_\xi = \Theta(\phi, \mathcal{L}_\xi \phi) - \xi \cdot L, \qquad dJ_\xi \simeq 0, \tag{41}$$

so that on shell

$$J_\xi \simeq dQ_\xi. \tag{42}$$

The Hamiltonian generator satisfies

$$\delta H_\xi = \int_{\partial\Sigma} \left(\delta Q_\xi - \xi \cdot \Theta \right). \tag{43}$$

Since $\Sigma \simeq S^3$ is boundaryless, $\partial\Sigma = \varnothing$, and pure-gauge diffeomorphism charges vanish globally:

$$H_\xi[\Sigma] = 0 \qquad (\xi \text{ gauge}). \tag{44}$$

This is the Noether consistency condition for a closed universe. Local or quasi-local charges can still be defined on finite subregions with an artificial boundary, but the total gauge generator on the full compact slice vanishes, as required by Noether's second theorem.

IX. DISCUSSION

A. Two-Scale Structure: $R_{\rm eff}$ vs Global Curvature

The inferred $R_{\rm eff} \approx 475\,\mathrm{Mpc}$ at $z = 0.8$ is *not* the global curvature radius of the Universe: the Planck constraint $|\Omega_k| < 5 \times 10^{-3}$ [9] sets $R_{\rm cosm}^{\rm global} > 10\,\mathrm{Gpc}$. The two-scale structure is as follows:

- *Global topological radius* $R_{\rm cosm}^{\rm global}$: the actual curvature radius of the cosmological 3-sphere, constrained by Planck to $> 10\,\mathrm{Gpc}$, governing the CMB power at low ℓ.
- *Effective shell curvature scale* $R_{\rm eff}$: the local spectral scale operative within a narrow redshift shell, inferred from ring matching to be 474.511 Mpc at $z = 0.8$.

The $R_{\rm eff}$ scale may arise from non-linear collapse and mode-selection dynamics within the redshift shell, i.e. as a dynamically selected coherence scale. A detailed derivation would require numerical simulations on S^3 (see Open Problems below).

B. Relation to Conformal Cyclic Cosmology

Penrose's Conformal Cyclic Cosmology (CCC) [13] offers an alternative mechanism for ring-like features: imprints from past-aeon black-hole collisions. The compact-S^3 projected-shell framework is complementary. Both frameworks predict ring-like structures; they differ in the predicted correlation between ring positions and CMB anomalies. CCC predicts correlation with Penrose circles [14]; our framework predicts enhanced power at compact projected-shell modes. Future data can distinguish the two.

C. Open Problems

(1) *Bayesian evidence*: Construct a covariance matrix for $(L_{\rm GA}, d_{\rm GR}, d_{\rm BR}, \lambda_c)$ and compute the evidence of the compact-S^3 projected-chord model relative to flat $\mathbb{R}^3$, periodic T^3, BAO-like, CCC-like, and string-like templates.
(2) *N-body simulations on S^3*: Numerical structure formation on S^3 with the appropriate boundary conditions (not periodic torus) has not been performed. StePS-like compactified simulations are a useful precedent, but they are not yet a direct nonlinear S^3 calculation.
(3) *Origin of the $R_{\rm eff}$ scale*: Why does structure formation at $z \approx 0.8$ select $R_{\rm eff} \approx 475\,\mathrm{Mpc}$?
(4) *Mode-selection rule*: The observed set $(1, 2, 6, 8)$ must be derived from dynamics, a survey-window selection rule, or a prior on the low-ℓ power spectrum before the model becomes predictive in a first-principles sense.
(5) *Same-field prediction test*: Deeper surveys should search the same redshift shell for coherent features near the additional projected diameters listed in Table II.
(6) *Interaction with CMB*: Do the same compact shell modes generate detectable signatures in the CMB covariance or low-ℓ power spectrum at an observable level?

X. CONCLUSIONS

This paper develops a compact-S^3 consistency analysis for the Lopez–Clowes ring-like field near $z \approx 0.8$. The robust analytic core is independent of any specific mode assignment: compactness gives a discrete global mode basis, forbids a pure exponential-tail intuition for globally supported correlations, and naturally allows shell-like structures whose narrow-slice projections appear as rings or arcs.

For the phenomenological size analysis, we use the exact projected-chord law

$$d_\ell = 2R_{\rm eff} \sin\left(\frac{\pi}{\ell + 1}\right). \tag{45}$$

Earlier Fibonacci scaling appears only as an asymptotic large-ℓ limit. With the empirical mode assignment $(1, 2, 6, 8)$ and the best-fit shell scale $R_{\rm eff} = 474.511\,\mathrm{Mpc}$, the model reproduces the four observed scales at the percent level.

The paper does not claim that compact topology has been observationally established, nor that the mode assignment has been derived from first principles. The narrower claim is that a compact-S^3 projected-shell framework now admits an exact geometric size law and provides a sharply testable alternative to flat-space intuition for this field. The Bayesian and same-field tests described above turn this consistency analysis into a falsifiable programme rather than a completed detection claim.

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AUTHOR CONTRIBUTIONS

A.P. and B.B. jointly developed the conceptual framework, analytical arguments, manuscript structure, and interpretation. Both authors reviewed and approved the submitted version.

DATA AVAILABILITY

No new observational or simulation datasets were generated in this study. All data discussed are drawn from published literature and public preprint sources cited in the reference list.

Appendix A: Reproducibility Skeleton for the Projected-Chord Fit

The fit in Table I is reproducible from the following minimal computation:

$$s_i = 2\sin\!\left(\frac{\pi}{\ell_i+1}\right), \qquad \widehat{R}_{\rm eff} = \frac{\sum_i d_i^{\rm obs} s_i}{\sum_i s_i^2}, \tag{A1}$$

with

$$(\ell_i) = (1,2,6,8), \qquad (d_i^{\rm obs}) = (940,840,400,320)\,{\rm Mpc}. \tag{A2}$$

This gives $\widehat{R}_{\rm eff} = 474.511\,{\rm Mpc}$. A Bayesian extension replaces the ordinary least-squares objective by Eq. (23); no Bayes factor is quoted here because that requires an observational covariance matrix not yet available from the published uLSS size estimates.

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